

BOW ISLAND, AB, Canada

WMO: 712310

Lat: **49.7344N**Lon: **111.4503W**Elev: **817**StdP: **91.89**Time Zone: **-7.00 (NAM)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-25.7	-32.9	0.2	-29.2	-29.1	0.3	-25.5	15.0	0.6	13.4	-0.1	3.1	160	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	31.9	18.0	30.0	17.3	28.1	16.9	20.1	28.3	19.0	27.0	17.9	25.6	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.7	23.8	16.0	12.6	22.6	14.9	11.7	21.5	61.5	28.2	57.4	27.1	53.8	25.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.9	9.5	-33.7	35.1	3.6	1.5	-36.3	36.2	-38.4	37.0	-40.4	37.9	-43.0	38.9		
(5)			-34.0	22.9	4.2	1.4	-37.1	23.9	-39.5	24.8	-41.9	25.6	-45.0	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-7.8	-6.4	-0.6	6.0	11.7	15.7	19.1	18.3	13.4	6.5	-1.2	-7.1
	DBStd	11.51	9.13	8.71	7.94	4.98	4.59	3.35	3.02	3.42	4.63	5.20	7.58	8.37
	HDD10.0	2532	553	458	331	135	35	2	0	1	18	130	338	531
	HDD18.3	4732	811	692	588	369	210	92	27	43	157	368	586	789
	CDD10.0	961	0	0	1	16	87	172	283	258	121	21	2	0
	CDD18.3	120	0	0	0	0	4	13	51	42	10	0	0	0
	CDH23.3	2067	0	0	1	9	115	219	771	702	241	9	0	0
	CDH26.7	708	0	0	0	1	26	54	279	266	81	1	0	0
Wind	WSAvg	4.4	4.9	4.6	4.9	5.0	4.8	4.1	3.3	3.3	3.8	4.5	4.6	4.8
Precipitation	PrecAvg	364	17	13	19	30	47	83	36	34	31	21	15	14
	PrecMax	532	44	29	50	68	102	207	125	80	68	70	32	32
	PrecMin	193	6	2	1	7	7	15	3	2	0	6	2	0
	PrecStd	76	10	7	12	15	24	41	28	21	20	14	9	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	13.1	18.4	24.6	29.7	31.3	34.0	33.9	32.4	24.7	17.7	11.1
		MCWB	6.2	6.2	9.2	11.7	14.9	18.4	19.6	17.4	15.6	12.6	9.0	5.2
	2%	DB	7.6	9.1	14.8	21.2	26.6	28.1	31.8	31.8	29.1	20.7	13.7	7.7
		MCWB	3.9	4.0	7.5	10.5	13.7	16.8	19.5	17.1	15.0	11.4	7.1	3.6
	5%	DB	5.4	6.4	12.2	18.2	23.7	25.9	29.8	29.7	26.1	17.5	10.8	5.6
		MCWB	2.5	2.7	6.2	8.9	12.6	16.0	18.7	16.8	14.2	9.7	5.8	2.2
	10%	DB	3.9	4.2	9.2	15.2	21.1	23.6	27.8	27.5	23.0	14.8	8.3	3.8
		MCWB	1.5	1.5	4.4	7.4	11.3	14.8	18.2	16.5	12.8	8.2	4.0	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	7.0	9.7	12.7	16.1	20.2	22.9	20.4	17.5	13.4	10.0	6.0
		MCDB	9.9	12.6	17.2	22.3	26.0	28.4	30.3	29.2	27.6	22.7	16.3	10.0
	2%	WB	4.1	4.6	8.1	10.9	14.6	18.3	20.9	19.1	15.8	11.5	7.6	4.0
		MCDB	7.3	8.7	13.7	20.1	24.2	25.5	28.9	27.8	26.0	19.0	13.0	7.3
	5%	WB	2.6	2.9	6.3	9.5	13.4	16.9	19.7	18.1	14.6	10.0	6.0	2.5
		MCDB	5.1	6.0	11.6	17.3	22.3	24.1	27.7	26.6	24.2	16.3	10.6	5.4
	10%	WB	1.6	1.6	4.5	7.9	12.2	15.6	18.6	17.1	13.5	8.6	4.4	1.1
		MCDB	3.8	4.3	8.8	14.1	20.0	22.2	26.3	25.4	22.5	14.1	8.2	3.8
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.2	11.4	12.9	13.2	12.5	14.9	15.3	14.5	12.6	10.9	10.2
		MCDBR	10.7	12.9	15.3	18.4	17.9	16.5	18.4	19.7	20.2	16.7	13.9	11.0
		MCWBR	8.0	8.8	9.1	9.4	8.0	7.7	9.0	8.4	8.8	8.8	8.6	7.8
	5% WB	MCDBR	10.6	12.6	14.6	17.0	15.8	14.9	16.8	16.7	18.4	15.7	13.2	10.9
		MCWBR	8.1	8.9	9.0	9.1	7.5	7.7	8.9	8.0	8.4	8.7	8.5	8.0
Clear-Sky Solar Irradiance	taub	0.265	0.270	0.292	0.341	0.362	0.354	0.354	0.375	0.333	0.292	0.264	0.255	
	taud	2.430	2.476	2.472	2.330	2.317	2.394	2.407	2.319	2.412	2.512	2.498	2.418	
	Ebn at Noon	804	895	927	904	895	904	898	859	865	853	805	764	
	Edh at Noon	60	74	89	115	122	114	111	115	94	71	56	53	
All-Sky Solar Radiation	RadAvg	1.26	2.26	3.58	4.94	5.85	6.44	7.01	5.63	4.01	2.50	1.41	0.99	
	RadStd	0.11	0.18	0.25	0.34	0.42	0.38	0.35	0.36	0.33	0.25	0.09	0.08	